

BI Strategy Note

Section 1 – Context

The proposed Business Intelligence (BI) strategy focuses on HimShakti's **Food Processing Raw Material Expiry & Inventory Waste Dashboard**. The objective is to help warehouse managers, inventory supervisors, and production teams monitor raw material inventory, track batch expiry dates, reduce inventory waste, and improve production planning through data-driven decision-making. This strategy supports efficient inventory management while minimizing losses caused by expired raw materials.

Section 2 – Recommended KPIs

The following Key Performance Indicators (KPIs) are recommended to monitor inventory performance and reduce expiry-related losses.

1. Total Inventory Quantity

Measures the total quantity of raw materials currently available in the warehouse.

2. Remaining Inventory Quantity

Tracks the remaining stock available after production to support inventory planning.

3. Expiring Soon Batches

Identifies batches that will expire within the next 7–15 days to reduce waste.

4. Average Shelf Life

Calculates the average shelf life of finished goods for better production scheduling.

5. Monthly Batch Entries

Monitors the number of raw material batches received every month to understand procurement trends.

6. Inventory Utilization Rate

Measures how efficiently available inventory is consumed during production.

7. Inventory Waste Percentage

Shows the percentage of inventory lost due to expiry or spoilage and helps monitor operational efficiency.

Section 3 – Data Source

The dashboard will use multiple business data sources to generate meaningful insights. Raw material information such as Batch ID, Entry Date, Expiry Date, Quantity, and Remaining Quantity will be collected from warehouse inventory logs. Production Date and Finished Goods details will be obtained from production records. Procurement records will provide supplier and purchasing information, while daily inventory updates can be maintained using Google Sheets or Excel. The cleaned dataset will be stored in CSV format and processed using Python before being connected to the BI dashboard.

Section 4 – Recommended Tool Stack

The recommended tool stack uses free and industry-standard technologies. Google Forms or Excel can be used for daily inventory data collection. Google Sheets and CSV files will serve as the primary data storage. Python with the Pandas library will handle data cleaning and preparation. Google Looker Studio will be used to build interactive dashboards for visualization and reporting. GitHub will be used for version control and project management. ChatGPT can assist in generating business insights, documentation, and analytical recommendations, making the overall BI workflow more efficient.

Section 5 – Phased Implementation Plan

The implementation should follow a phased approach. In Phase 1, establish a standard process for collecting daily inventory and production data. Phase 2 focuses on cleaning and validating the collected data using Python before storing it in a structured CSV format. Phase 3 involves developing an interactive dashboard in Looker Studio with KPI cards, charts, and filters. Finally, Phase 4 uses dashboard insights to support inventory planning, minimize expiry-related losses, and improve operational decision-making across the organization.

Section 6 – BI Case Study 1 (Swiggy)

Company: Swiggy

Business Problem :-

Swiggy needed to improve delivery efficiency, reduce delivery time, optimize order allocation, and enhance customer satisfaction while managing a rapidly growing user base.

KPIs Tracked:

- Average Delivery Time
- Order Completion Rate
- Customer Satisfaction Score
- Delivery Partner Utilization
- Order Cancellation Rate

- Daily Active Users

Data Sources:

- Customer Orders
- Delivery Partner GPS
- Restaurant Data
- Customer Ratings
- Payment Transactions

Tool Stack:

- Python
- Apache Spark
- AWS
- Redshift
- Tableau
- Airflow

Business Outcome:

Using Business Intelligence and analytics, Swiggy optimized delivery routes, reduced delivery delays, improved partner utilization, and increased customer satisfaction. Real-time dashboards enabled faster operational decisions and better resource allocation.

Source:

<https://bytes.swiggy.com/>

Section 7 – BI Case Study 2 (Zomato)

Company: Zomato

Business Problem:

Zomato needed to improve restaurant recommendations, optimize delivery operations, monitor customer engagement, and reduce order cancellations across different cities.

KPIs Tracked:

- Daily Orders
- Average Delivery Time
- Customer Ratings
- Restaurant Performance
- Order Cancellation Rate
- Revenue Growth

Data Sources:

- Customer Orders

- Restaurant Database
- Delivery Data
- Payment Records
- Customer Feedback

Tool Stack:

- Python
- SQL
- BigQuery
- Tableau
- Machine Learning Models

Business Outcome:

Zomato used Business Intelligence dashboards to monitor business performance, improve restaurant recommendations, optimize delivery operations, and support strategic business decisions using real-time analytics.

Source:

<https://www.zomato.com/blog>

Section 8 – BI Maturity Model Mapping

Dimension	Current Stage	Justification
Data Availability	Developing	Inventory and production data are collected but require better integration and standardization.
Tooling	Developing	Basic spreadsheets and dashboards are available, but advanced BI tools are still being adopted.
Team	Emerging	Inventory analysis is handled by operational staff with limited dedicated BI resources.
Use of Insights	Developing	Business decisions are partially data-driven, but predictive analytics and automation are yet to be implemented.

Overall BI Maturity Level: Developing

Justification

HimShakti currently falls under the **Developing** stage of BI maturity. The organization collects essential inventory and production data but lacks an integrated Business Intelligence ecosystem. Basic reporting and dashboarding can support operational decisions; however, advanced analytics, automated alerts, predictive inventory planning, and a dedicated BI team are still required. Implementing modern BI tools and standardized data collection processes will help the organization progress toward an Established BI maturity level.

